Politecnico di Milano
Dipartimento di Elettronica, Informazione e Bioingegneria
prof. Fabrizio Ferrandi

Parallel Computing–I part–Thursday, August 30th, 2024

Polimi ID ________ ____________________________

Surname ________ ______________________ **Name** _________________________

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _________________________

Exercise 2 (4 points) _________________________

Exercise 3 (4 points) _________________________

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Exercise n. 1

Answer the following questions about PRAM models and briefly explain (without an explanation, the answer will be considered invalid)

A. In a PRAM model, each processor has an unbounded number of registers. What purposes could such registers be used for? Is the access time different from the one for the global shared memory? (1)

B. Please discuss the lemma that describes what occurs when we have a number of processors M' less than the one required by the PRAM model. (1)

C. Please define the Cost in a PRAM model. (1)

D. Could you please discuss the following sentence? "Gustafson's law presupposes that the computing requirements will stay the same, given increased processing power. In other words, an analysis of the same data will take less time given more computing power". Is it true or false? What does it mean? (1)

Exercise n. 2

Answer the following questions about different forms of parallel executions and briefly explain (without an explanation, the answer will be considered invalid).

A. Please describe the key characteristics of data-parallel program execution with SIMD instructions in the presence of conditional instructions. (2)

B. Explain how hardware-supported multi-threading. (2)

Exercise n. 3

Answer the following questions about programming models plus CUDA and briefly explain (without an explanation, the answer will be considered invalid).

A. Please describe what is meant by **"CUDA thread blocks"**. (1)

B. Please describe the organization of memory in a GPU system and discuss the different memory latencies. (2)

C. What is meant for the Block/Tiling CUDA code optimization technique? (1)

Exercise n. 4

Answer the following questions about memory and heterogeneous systems and briefly explain (without an explanation, the answer will be considered invalid).

A. Please briefly describe when coalesced memory access happened. (1) **This question can be skipped in case of a passed challenge.**

B. Please explain the conditions under which a memory system is defined as “consistent”. Which of the conditions a **sequentially consistent memory system** is relaxing? (1).

C. Please briefly describe what DRAM Banking is meant for. (1)

D. What do the terms "custom computing" and "heterogeneous systems" mean? (1)